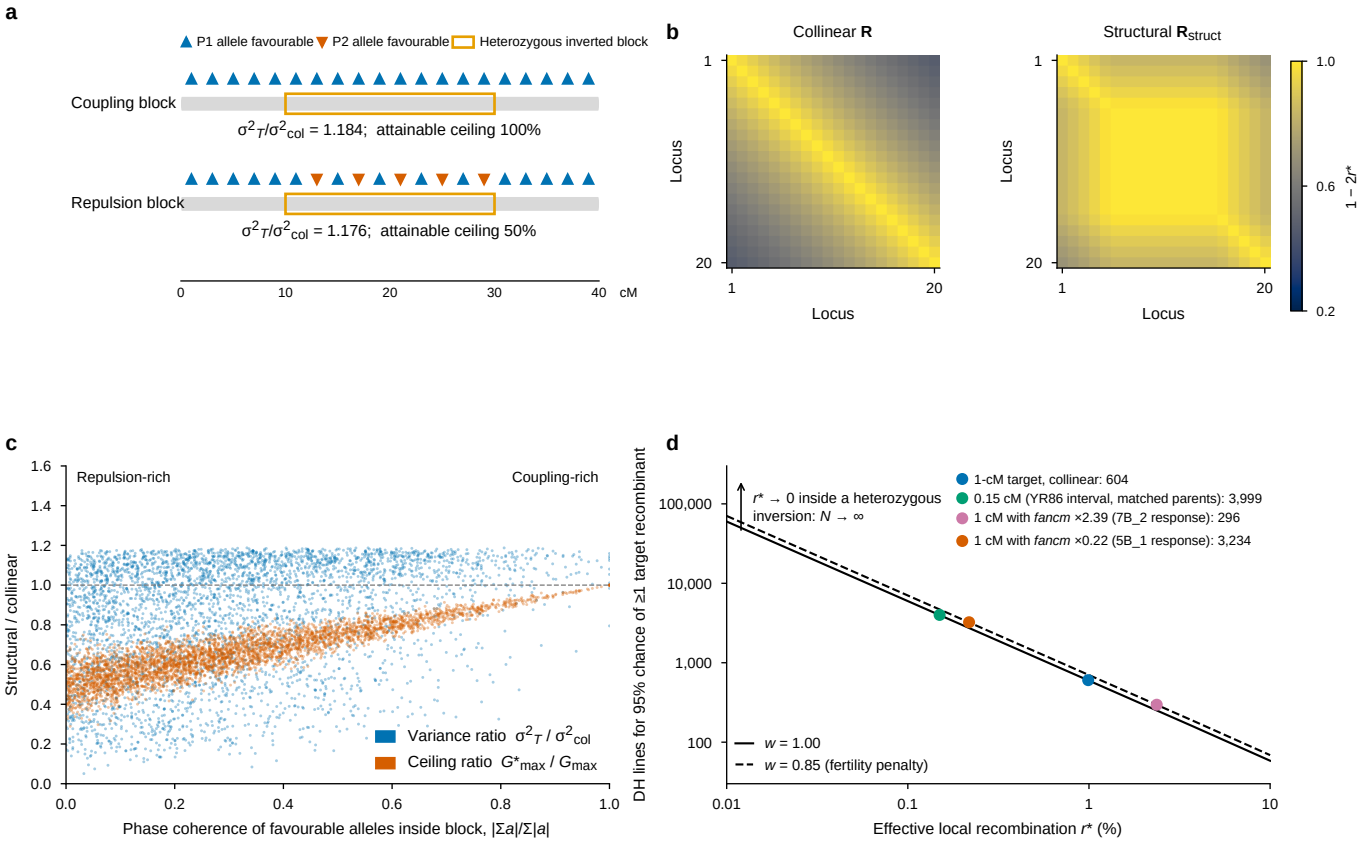

a
P1 allele favourable
P2 allele favourable
Heterozygous inverted block
Coupling block
σ²T/σ²col = 1.184; attainable ceiling 100%
Repulsion block
σ²T/σ²col = 1.176; attainable ceiling 50%
0
10
20
30
40
cM
b
Collinear R
Structural Rstruct
Locus
1
20
1 − 2r*
1.0
0.6
0.2
c
Repulsion-rich
Coupling-rich
Structural / collinear
Phase coherence of favourable alleles inside block, |Σa|/Σ|a|
Variance ratio σ²T / σ²col
Ceiling ratio G*max / Gmax
d
r* → 0 inside a heterozygous inversion: N → ∞
1-cM target, collinear: 604
0.15 cM (YR86 interval, matched parents): 3,999
1 cM with fancm ×2.39 (7B_2 response): 296
1 cM with fancm ×0.22 (5B_1 response): 3,234
w = 1.00
w = 0.85 (fertility penalty)
DH lines for 95% chance of ≥1 target recombinant
Effective local recombination r* (%)